Year 2 Pure Mathematics

Chapter 9

Differentiation

Practice Booklet

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9.1 Differentiating $\sin x$ and $\cos x$

Practice Book, printed page 70.

1. Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ given that:

a. $y = 6\sin x$

b. $y = -3\sin\frac{1}{2}x$

c. $y = \frac{1}{3}\sin 5x$

d. $y = \cos 4x$

e. $y = 5\cos 6x$

f. $y = -\frac{1}{2}\cos 7x$

Hint

If $y = \sin kx$, then $\frac{\mathrm{d}y}{\mathrm{d}x} = k\cos kx$

If $y = \cos kx$, then $\frac{\mathrm{d}y}{\mathrm{d}x} = -k\sin kx$

2. Find $\mathrm{f}'(x)$ given that:

a. $\mathrm{f}(x) = 4\sin x - 3\cos x$

b. $\mathrm{f}(x) = \frac{1}{2}\sin 2x + 4\cos x - 6\sin\frac{1}{4}x$

c. $\mathrm{f}(x) = 5x^{\frac{3}{2}} - \frac{4}{x^2} - 6\sin\frac{1}{3}x$

d. $\mathrm{f}(x) = \frac{8}{\sqrt{x}} + 3\cos\frac{1}{6}x$

Hint

Differentiate each term separately.

3. $f(x) = 5\sin x - 3\cos x$. Find the x-coordinates of the stationary points on the curve with equation $y = f(x)$ in the interval $-180° \leq x \leq 180°$.

Hint

Stationary points occur when the gradient is equal to zero.

4. $f(x) = x - \cos 2x$. Find the equation of the tangent to the curve at the point $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$.

Hint

First differentiate to find the gradient, m. Then find the equation of the tangent using the equation $y - y_1 = m(x - x_1)$
⟵ Year 1, Section 12.6

5. E $f(x) = 6\sin x - 3\cos 2x$
Find the stationary points of the curve $y = f(x)$ in the interval $0 \leq x \leq 2\pi$. (6 marks)

6. E/P $g(x) = 6\sin 2x - 4\cos 2x$. Find the equation of the normal to the curve with equation $y = g(x)$ at the point $(\pi, -4)$. (5 marks)

7. E/P $h(x) = 3\sin x - 4\cos x + \frac{1}{2}x^2$. Show that the equation of the tangent to the curve with equation $y = h(x)$ at the point with x-coordinate $\frac{\pi}{2}$ is $8y - (32 + 4\pi)x = 24 - 16\pi - \pi^2$. (6 marks)

8. E/P Prove from first principles that the derivative of $\cos x$ is $-\sin x$. (5 marks)

9.2 Differentiating exponentials and logarithms

Practice Book, printed page 70.

1. Differentiate:

a. $6e^{-3x}$

b. $5e^{2x} - 7e^{-x}$

c. $\frac{7}{e^{5x}}$

d. $\frac{(1 - e^{2x})^2}{e^{2x}}$

Hint

If $y = e^{kx}$, then $\frac{dy}{dx} = ke^{kx}$

2. Find $f'(x)$ given that:

a. $f(x) = \ln 2x$

b. $f(x) = \ln x^2$

c. $f(x) = \ln\left(\frac{7}{x}\right)$

d. $f(x) = \ln(4x^3)$

Hint

If $y = \ln x$, then $\frac{dy}{dx} = \frac{1}{x}$
First expand the expression using laws of logarithms. ⟵ Year 1, Section 14.5

3. Find $\frac{dy}{dx}$ given that:

a. $y = 3^x$

b. $y = 5^{2x}$

c. $y = \left(\frac{2}{5}\right)^{4x}$

d. 8^{-x}

Hint

If $y = a^{kx}$, where k is a real constant and $a > 0$, then $\frac{\mathrm{d}y}{\mathrm{d}x} = a^{kx} k \ln a$.

4. Find the equation of the tangent to the curve $y = 4^x$ at the point with x-coordinate $-\frac{1}{2}$

5. E Given $\mathrm{f}(x) = e^{-x} + 2\ln x^3$, find $\mathrm{f}'(x)$. (3 marks)

6. E/P A curve has equation $y = 3e^x - \ln x^2$. Show that the equation of the normal at the point with x-coordinate 1 is $x + (3e - 2)y - 9e^2 + 6e - 1 = 0$. (6 marks)

7. E/P $\mathrm{f}(x) = 4^x + 4^{-x}$. Show that the equation of the of the tangent to the curve with equation $y = \mathrm{f}(x)$ at the point with x-coordinate $\frac{1}{2}$ is $y = (\ln 8)x + \frac{5}{2} - \frac{1}{2}\ln 8$. (6 marks)

8. E/P $\mathrm{f}(x) = 9^{3x} + 3^{6x}$

a. Demonstrate, clearly showing all your steps, that $\mathrm{f}(x)$ can be written in the form $a(3^{kx})$ where a and k are constants to be found. (2 marks)

b. Hence, or otherwise, show that $\mathrm{f}'(x) = (4\ln 3)(3^{6x+1})$ (3 marks)

9.3 The chain rule

Practice Book, printed page 71.

1. Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ for each of the following:

a. $y = (4 - 3x)^3$

b. $y = (6 + 5x)^{\frac{1}{2}}$

c. $y = (5 + 2x^3)^4$

d. $y = (7 + 3x^2)^{-5}$

e. $y = \frac{6}{\sqrt{4 - 5x}}$

f. $y = \frac{1}{6 - x}$

g. $y = \sqrt[3]{3 + 2x^4}$

h. $y = \sqrt{7x^3 - 4}$

Hint

If $y = (\mathrm{f}(x))^n$, then $\frac{\mathrm{d}y}{\mathrm{d}x} = n(\mathrm{f}(x))^{n-1}\mathrm{f}'(x)$.

2. Differentiate:

a. $y = \sin^3 x$

b. $y = \ln x^4$

c. $y = e^{\cos 4x}$

d. $y = (\ln x)^2$

e. $y = \ln(\cos x)$

f. $y = \sin(e^{-3x})$

Hint

You can use the chain rule to differentiate a function of a function.
If $y = \mathrm{f}(\mathrm{g}(x))$, then $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{f}'(\mathrm{g}(x))\mathrm{g}'(x)$.

3. $\mathrm{f}(x) = (3 - \ln 6x)^2$

a. Find $\mathrm{f}'(x)$.

b. Evaluate $\mathrm{f}'\left(\dfrac{e^2}{6}\right)$.

4. Find the value of $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ at the point $(3, 6)$ on the curve with equation $2y^2 - 4y = x$.

Hint

First find $\dfrac{\mathrm{d}x}{\mathrm{d}y}$.
Then use the fact that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{1}{\frac{\mathrm{d}x}{\mathrm{d}y}}$

5. E/P A curve has equation $y = (2x - 3)^5$. Find the equation of the normal to the curve at the point P with x-coordinate 2. (5 marks)

6. E/P $\mathrm{f}(x) = \dfrac{3}{(6 - x)^3}$, $x \neq 6$. Find the tangent to the curve with equation $y = \mathrm{f}(x)$ at the point P with x-coordinate 4. Give your answer in the form $ax + by + c = 0$. (7 marks)

7. E/P $\mathrm{f}(x) = 3\sin^2 x - 2\cos^2 x$. Show that the equation of the tangent to the curve with equation $y = \mathrm{f}(x)$ at the point Q with x-coordinate $-\dfrac{5\pi}{6}$ is $12y - 30\sqrt{3}x - 25\pi\sqrt{3} + 9 = 0$. (6 marks)

8. E/P The curve C has equation $x = -2\sin 4y$. Show that the equation of the normal to C at the point A with y-coordinate $\dfrac{2\pi}{3}$ is $4x + y + 4\sqrt{3} - \dfrac{2\pi}{3} = 0$. (6 marks)

9. E/P Find the equation of the tangent to the curve with equation $y = 2^{(x^4)}$ at the point with coordinates $(-1, 2)$. Give your answer in the form $ax + by + c = 0$, where a, b and c are exact values to be found. (6 marks)

9.4 The product rule

Practice Book, printed page 72.

1. Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ for each of the following:

a. $y = x(4 - 5x)^3$

b. $y = 3x^3(2x - 1)^6$

c. $y = \dfrac{4}{x^2}(6x - 7)^4$

d. $y = 8x^2(2x^3 - 4)^{-3}$

Hint

If $y = uv$, then $\dfrac{\mathrm{d}y}{\mathrm{d}x} = u\dfrac{\mathrm{d}v}{\mathrm{d}x} + v\dfrac{\mathrm{d}u}{\mathrm{d}x}$, where u and v are functions of x.

2. Find $\mathrm{f}'(x)$ given that:

a. $\mathrm{f}(x) = x^2\cos 2x$

b. $\mathrm{f}(x) = e^{3x}\sin 5x$

c. $\mathrm{f}(x) = e^{-x}(2x^2 + 3)^3$

d. $\mathrm{f}(x) = -3\cos 4x \sin 2x$

e. $\mathrm{f}(x) = x^2 \ln x$

f. $\mathrm{f}(x) = e^{4x} \ln(\cos 3x)$

Hint

If $\mathrm{f}(x) = \mathrm{g}(x)\mathrm{h}(x)$ then
$\mathrm{f}'(x) = \mathrm{g}(x)\mathrm{h}'(x) + \mathrm{h}(x)\mathrm{g}'(x)$

3. Find the value of $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ at the point $(1, 4)$ on the curve with equation $y = 4x^3(1 - 2x^2)^4$.

Hint

Evaluate $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ when $x = 1$.

4. Differentiate with respect to x:

a. $(x^4 - 3x^2)e^{-2x}$

b. $(4\cos 2x - \sin x)e^{3x}$

c. $\left(x^2 - 5\cos\dfrac{1}{2}x\right)\ln 4x$

5. E Differentiate with respect to x:

a. $e^{-x}\cos 3x$ (2 marks)

b. $x^4 \ln(\sin x)$ (2 marks)

6. E/P

a. A curve C has equation $y = 2x^4(2 - 5x^2)^3$. Show that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = x^3(2 - 5x)^n(ax^2 + bx + c)$, where n, a, b and c are constants to be found. (4 marks)

b. Hence find the exact x-coordinates of the stationary points of C. (4 marks)

7. E/P The curve C has equation $\mathrm{f}(x) = e^{4x}\sin x$.

a. Show that the turning points of C occur when $\tan x = -\dfrac{1}{4}$ (3 marks)

b. Show that the equation of the normal to C at the point where $x = \dfrac{\pi}{2}$ is
$x + 4e^{2\pi}y - 4e^{4\pi} - \dfrac{\pi}{2} = 0$ (3 marks)

8. E/P A curve has equation $y = (x - \pi)\sin 2x$. Show that the equation of the tangent to the curve at the point with x-coordinate $\dfrac{3\pi}{4}$ is $x + y = \pi$. (6 marks)

9.5 The quotient rule

Practice Book, printed page 73.

1. Find $\mathrm{f}'(x)$ given that:

a. $\mathrm{f}(x) = \dfrac{4x + 3}{2x - 9}$

b. $\mathrm{f}(x) = \dfrac{6x^2}{3 - 5x}$

c. $\mathrm{f}(x) = \dfrac{1 - 8x}{(3x^3 + 2)^2}$

d. $\mathrm{f}(x) = \dfrac{7x^2 - 9x}{\sqrt{x^2 + 6}}$

Hint

If $f(x) = \dfrac{g(x)}{h(x)}$ then
$f'(x) = \dfrac{h(x)g'(x) - g(x)h'(x)}{(h(x))^2}$

2. Find $\dfrac{dy}{dx}$ for each of the following:

a. $y = \dfrac{4x^2}{\cos 2x}$

b. $y = \dfrac{x - e^{-x}}{\ln x}$

c. $y = \dfrac{e^x}{\cos^2 x}$

d. $y = \dfrac{5 - e^{-x}}{\sin 4x}$

Hint

If $y = \dfrac{u}{v}$ then $\dfrac{dy}{dx} = \dfrac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$ where u and v are functions of x.

3. Find the gradient of the curve with equation $y = \dfrac{8 - 2x}{4x + 1}$ at the point $\left(\dfrac{1}{2}, \dfrac{7}{3}\right)$.

4. The curve C has equation $y = \dfrac{x^2 - 6x}{e^{2x}}$

a. Find $\dfrac{dy}{dx}$.

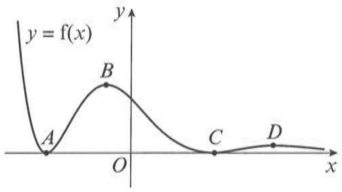
b. Find the equation of the normal to C at $\left(1, -\dfrac{5}{e^2}\right)$.

5. E Given $y = \dfrac{\sin^2 x}{e^{2x}}$, find $\dfrac{dy}{dx}$ (4 marks)

6. E Differentiate $\dfrac{e^{4x}}{x(x-5)}$ with respect to x. (4 marks)

7. E/P The curve C has equation $x = \dfrac{\cos 2y}{e^y}$. Find the equation of the tangent to the curve at the point where $y = \dfrac{\pi}{8}$ in the form $ax + by + c = 0$. (6 marks)

8. E/P The diagram shows part of the graph with equation $y = f(x)$, where $f(x) = \dfrac{\cos^2 x}{e^x}$



a. Show that at the stationary points A, B, C and D, either $\cos x = 0$ or $\tan x = -\dfrac{1}{2}$ (4 marks)

b. Hence, or otherwise, find the x-coordinates of the points A, B, C and D, giving your answers in radians to 2 decimal places. (2 marks)

9.6 Differentiating trigonometric functions

Practice Book, printed page 74.

1. Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ given:

a. $y = \tan 4x$

b. $y = 7\cot 2x$

c. $y = \cot^5 x$

d. $y = -3\tan^2 x$

Hint

If $y = \tan kx$, then $\dfrac{\mathrm{d}y}{\mathrm{d}x} = k\sec^2 kx$

If $y = \cot kx$, then $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -k\,\mathrm{cosec}^2 kx$

2. Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ given:

a. $y = \mathrm{cosec}\, 6x$

b. $y = -4\sec 5x$

c. $y = (\mathrm{cosec}\, x)^{\frac{4}{3}}$

d. $y = -2\sec^3 4x$

Hint

If $y = \mathrm{cosec}\, kx$, then $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -k\,\mathrm{cosec}\, kx \cot kx$

If $y = \sec kx$, then $\dfrac{\mathrm{d}y}{\mathrm{d}x} = k\sec kx \tan kx$

3. Find $\mathrm{f}'(x)$ when $\mathrm{f}(x)$ is:

a. $x^2 \tan 3x$

b. $\dfrac{e^{\sec x}}{x^2+1}$

c. $\ln(\mathrm{cosec}^2 x)$

d. $e^{-x}\cot 4x$

e. $\dfrac{\sec^2 x}{\ln x}$

f. $\dfrac{(4x-1)^3}{\cot^2 x}$

Hint

Use the product rule or quotient rule in addition to the chain rule and standard rules for differentiating trigonometric functions.

4. Given that $x = \sin 2y$,

a. find $\dfrac{\mathrm{d}x}{\mathrm{d}y}$ in terms of y.

b. Hence find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of x.

Hint

Use the identity $\sin^2 A + \cos^2 A \equiv 1$ to write $\cos 2y$ in terms of x.

⟵ Year 1, Section 10.3

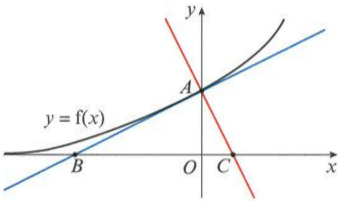
5. E Show that if $y = \mathrm{cosec}\, x$, then $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -\,\mathrm{cosec}\, x \cot x$. (5 marks)

6. E/P $\mathrm{f}(x) = \ln(\cot 2x)$

a. Show that $\mathrm{f}'(x) = -4\operatorname{cosec} 4x$. (5 marks)

b. Find the equation of the normal to the curve $y = \mathrm{f}(x)$ at the point with x-coordinate $\frac{\pi}{8}$ (3 marks)

7. E/P The diagram shows part of the curve $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = e^{\tan \frac{1}{2}x}$



Point A is the y-intercept of $y = \mathrm{f}(x)$. The tangent and normal to the curve $y = \mathrm{f}(x)$ at A intersect the x-axis at the points B and C respectively.
Find the area of triangle ABC. (8 marks)

8. E/P Given that $x = \sec 4y$,

a. find $\dfrac{\mathrm{d}x}{\mathrm{d}y}$ in terms of y. (2 marks)

b. Hence find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of x. (4 marks)

9.7 Parametric differentiation

Practice Book, printed page 75.

1. Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ for each of the following, leaving your answers in terms of the parameter t.

a. $x = 4t - 1$, $y = 2t^3$

b. $x = \dfrac{2}{t^2}$, $y = 4 - t^2$

c. $x = 3t$, $y = te^{4t}$

d. $x = 4\sin t$, $y = 5\tan t$

e. $x = 4e^{3t} - 1$, $y = 2e^t + 4$

f. $x = 2t^3 - 1$, $y = \ln t$

Hint

If x and y are given as functions of a parameter, t, then $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\frac{\mathrm{d}y}{\mathrm{d}t}}{\frac{\mathrm{d}x}{\mathrm{d}t}}$

2. A curve C has parametric equations $x = \tan t$, $y = \sec t$. Find:

a. $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of the parameter t

b. the equation of the tangent to the curve at the point P, where $t = \dfrac{\pi}{4}$

Hint

Substitute $t = \dfrac{\pi}{4}$ into the original equations to find the coordinates of P.

3. A curve C has parametric equations $x = 1 - t^2$, $y = t^3 + 2t^2$. Find:

a. $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of the parameter t

b. the equation of the normal to the curve at the point P, where $t = -3$.
Give your answer in the form $ax + by + c = 0$, where a, b and c are constants to be found.

4. A curve C has parametric equations $x = 1 - \cos 2\theta$, $y = 4\sin 2\theta$, $-\pi \leq \theta \leq \pi$. Find:

a. $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of the parameter θ

b. the coordinates of the points with zero gradient on the curve C.

Hint

Solve $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$ to find the values of θ at the points with zero gradient, then substitute into the original parametric equations to find the coordinates.

5. E/P The curve C has parametric equations $x = 2\sec\theta$, $y = 4\cos^2\theta$, $-\pi \leq \theta \leq \pi$.

a. Find an expression for $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of the parameter θ. (4 marks)

b. Find an equation of the normal to the curve at the point where $\theta = \dfrac{\pi}{3}$ (4 marks)

6. E/P The curve C has parametric equations $x = (t-2)^2$, $y = t^2 + 4t$, $t \in \mathbb{R}$.
The line l is a tangent to C and is parallel to the line with equation $y = 2x - 9$.
Find the equation of l. (6 marks)

7. E/P The curve C has parametric equations $x = -2\sec 2t$, $y = 2\cot t$, $-\dfrac{\pi}{2} \leq \theta \leq \dfrac{\pi}{2}$

a. Find an expression for $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of the parameter t. (4 marks)

The point A with coordinates $(-4, -2\sqrt{3})$ lies on the curve C. The line l is tangent to C at A.

b. Show that an equation for l is $x\sqrt{3} + 3y + 10\sqrt{3} = 0$. (4 marks)

8. E/P The curve C has parametric equations $x = \dfrac{t}{1-t}$, $y = \dfrac{1}{1+t^2}$, $-1 < t < 1$.

The line l is normal to C at the point where $t = -\dfrac{1}{2}$

a. Find an equation for the line l. (6 marks)

b. Show that a Cartesian equation for the curve C is $y = \dfrac{x^2 + 2x + 1}{2x^2 + 2x + 1}$ (4 marks)

9.8 Implicit differentiation

Practice Book, printed page 76.

1. By writing $u = y^5$ and using the chain rule, show that $\dfrac{\mathrm{d}}{\mathrm{d}x}(y^5) = 5y^4\dfrac{\mathrm{d}y}{\mathrm{d}x}$

Hint

Use $\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{\mathrm{d}u}{\mathrm{d}y} \times \dfrac{\mathrm{d}y}{\mathrm{d}x}$

2. Show that $\dfrac{\mathrm{d}}{\mathrm{d}x}(x^3y^2) = 2x^3y\dfrac{\mathrm{d}y}{\mathrm{d}x} + 3x^2y^2$

Hint

Use the product rule and the fact that $\dfrac{\mathrm{d}}{\mathrm{d}x}(y^n) = ny^{n-1}\dfrac{\mathrm{d}y}{\mathrm{d}x}$

3. Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of x and y where:

a. $x^4 - y^3 = 3$

b. $3x^2 + 4y^2 - 5x^2y = 2$

c. $y = \dfrac{5y^2}{x^2 - y^2}$

d. $4\ln(x-5) - \ln(y^3) = 0$

e. $\cos x + \sin^2 y = 1$

f. $xe^{-y} + ye^{2x} - 7 = 0$

Hint

When you differentiate implicitly, your expression for $\frac{dy}{dx}$ will usually involve both x and y.

4. The curve C is described by the equation

$$x^2 - 4x^2y + y^2 = 21$$

a. Find $\frac{dy}{dx}$.

b. Evaluate $\frac{dy}{dx}$ at the point $(2, -1)$.

c. Find an equation of the tangent to C at the point $(2, -1)$.

Hint

For part b, substitute $x = 2$ and $y = -1$ into your expression for $\frac{dy}{dx}$.

5. E Find the gradient of the curve with equation $3x^2 - y^3 - 5xy = 1$ at the point $(2, 1)$. (5 marks)

6. E Given that $e^{-3y} - e^{-2x} = 4xy$, find $\frac{dy}{dx}$ in terms of x and y. (5 marks)

7. E/P The curve C satisfies $\cos 2x + \sin 2y = 1$, where $-\frac{\pi}{4} < x < \frac{\pi}{4}$ and $0 < y < \frac{\pi}{6}$

a. Find an expression for $\frac{dy}{dx}$ (3 marks)

b. Find an equation of the tangent to C at the point P with x-coordinate $-\frac{\pi}{6}$, giving your answer in the form $ax + by + c = 0$, where a, b and c are constants to be found. (5 marks)

8. E/P A curve C is described by the equation $4x^2 + y^2 + 6x - 8y - 10 = 0$.
Find an equation of the normal to C at the point $(1, 8)$, giving your answer in the form $ax+by+c = 0$, where a, b and c are constants to be found. (7 marks)

9.9 Using second derivatives

Practice Book, printed page 77.

1. $f(x) = e^{-x}\cos 2x$

a. Find $f'(x)$.

b. Find $f''(x)$.

Hint

To find $f''(x)$, differentiate $f'(x)$.

2. $f(x) = 8x^2 + 3e^{4x}$

a. Find $f''(x)$.

b. Hence show that $f(x)$ is a convex function for all real values of x.

Hint

The function $f(x)$ is convex on the interval $[a, b]$ if and only if $f''(x) \geq 0$ for every value of x in that interval.

3. $f(x) = \cos 2x$, $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

a. Find $f''(x)$.

b. Find the range of values for which $f(x)$ is concave.

Hint

The function $f(x)$ is concave on the interval $[a, b]$ if and only if $f''(x) \leq 0$ for every value of x in that interval.

4. $f(x) = x(2x-1)^5$

a. Find $f'(x)$ and $f''(x)$.

b. Show that $f''(x) = 0$ at $x = \frac{1}{6}$ and $x = \frac{1}{2}$

c. Show that $x = \frac{1}{6}$ is a point of inflection by demonstrating that $f''(0.1)$ and $f''(0.2)$ have different signs.

Hint

A point of inflection is a point at which $f''(x)$ changes sign. To find the point of inflection, solve $f''(x) = 0$, then check that $f''(x)$ has different signs on either side of that point.

5. E Find the interval on which the function $f(x) = x^3 - 8x^2 + 6x - 2$ is convex. (4 marks)

6. E $f(x) = x(2x-5)^4$

a. Find $f'(x)$ and $f''(x)$. (4 marks)

b. Show that $f(x)$ has a point of inflection at $x = 1$. (3 marks)

7. E/P The curve C has equation $y = f(x)$ where $f(x) = \frac{\sin 4x}{e^{2x}}$, $0 \leq x \leq \frac{\pi}{4}$

Determine the range of values for which $f(x)$ is convex. Give your answer to 2 decimal places where necessary. (6 marks)

8. E/P $f(x) = x^3 e^x$. Find the exact range of values for which the curve C with equation $y = f(x)$ is concave. (6 marks)

9.10 Rates of change

Practice Book, printed page 77.

1. Given that $S = 6x^2$ and that $\frac{dx}{dt} = \frac{2}{3}$,

a. find $\frac{dS}{dt}$

b. evaluate $\frac{dS}{dt}$ when $x = 3$.

Hint

You can use the chain rule to connect rates of change in situations involving two or more variables. Here $\frac{dS}{dt} = \frac{dS}{dx} \times \frac{dx}{dt}$.

2. Given that $y = x\cos 4x$ and that $\frac{dx}{dt} = 3$, find $\frac{dy}{dt}$ when $x = \frac{5\pi}{16}$

3. Given that $y = \frac{e^{2x}}{x^2}$ and that $\frac{dx}{dt} = -\frac{1}{2}$, find $\frac{dy}{dt}$ when $x = 4$.

4. The volume of a sphere $V\,\text{cm}^3$ is related to its radius by the formula $V = \frac{4}{3}\pi r^3$.

a. Find $\frac{dV}{dr}$

b. Given that the rate of change of the radius in $\text{cm}\,\text{s}^{-1}$ is given by $\frac{dr}{dt} = 3$, find $\frac{dV}{dt}$

Hint

For part b, r varies with t, so you cannot treat it as a constant when you differentiate V with respect to t. You need to use the chain rule:
$\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{\mathrm{d}V}{\mathrm{d}r} \times \frac{\mathrm{d}r}{\mathrm{d}t}$

5. E At any given time, the rate of population growth in a small town is modelled as being proportional to the current population, P.

a. Write a differential equation to describe this model. (1 mark)

b. Explain how this model could be adapted if the population is declining at a rate proportional to its size. (1 mark)

6. E At time t years after monitoring began, the area covered by Arctic ice is $I\,\mathrm{km}^2$. The rate of decay of ice coverage is directly proportional to the square root of the existing area covered. Write down a differential equation in terms of I and t. (2 marks)

7. E The value, V in £s, of a car t years after it is purchased is modelled by the equation

$$V = 23500e^{-0.25t}, \quad t \geq 0$$

a. Find the value of the car 7 years after it was purchased. (1 mark)

b. Find the rate of change of value of the car after 10 years. (3 marks)

8. E The volume, $V\,\mathrm{cm}^3$, of a sphere is given by $V = \frac{4}{3}\pi r^3$ and the surface area, $S\,\mathrm{cm}^2$, of a sphere is given by $S = 4\pi r^2$.

a. Show that $V = \frac{1}{6\sqrt{\pi}}S^{\frac{3}{2}}$ (2 marks)

The surface area of the sphere is expanding at a constant rate of $4\,\mathrm{cm\,s}^{-1}$.

b. Show that $\frac{\mathrm{d}V}{\mathrm{d}t} = \sqrt{\frac{S}{\pi}}$ (2 marks)

Problem solving: Set A

Bronze

The curve C has equation $y = x^4 - \frac{20}{3}x^3 + 14x^2 + 6x - 4$.

a. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ (3 marks)

b. Show that the curve C is concave in the interval $\left[1, \frac{7}{3}\right]$. (3 marks)

Silver

The curve C has equation $y = 4x^3 \ln x^2$, $x > 0$.
Show that the curve C is convex when $x \geq e^{-\frac{5}{6}}$. (6 marks)

Gold

$f(x) = ax^4 + 2x^3 + 3x^2 - 10x + 5$. Given that $f(x)$ is convex for all real values of x, find the range of possible values of a. (6 marks)

Problem solving: Set B

Bronze

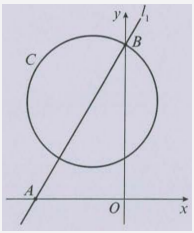
The curve C is described by $x^2 + y = (2 - x)(y - 1)$.

a. Find $\frac{dy}{dx}$ (4 marks)

b. Find the equation of the normal to C at the point with x-coordinate 0. (4 marks)

Silver

Circle C has parametric equations
$x = 2\sin t - 1$, $y = 2\cos t + 3$, $0 \leq t \leq 2\pi$
The line l_1 is normal to C at the point B, which lies on the y-axis.



a. Find the Cartesian equation of the circle, clearly stating the centre and radius. (3 marks)

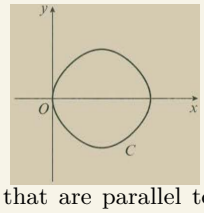
b. Show clearly that B has coordinates $(0, 3 + \sqrt{3})$. (2 marks)

The line l_1 intersects the x-axis at the point A.

c. Find the area of triangle AOB. (7 marks)

Gold

The diagram shows the curve C given by the equation
$\cos y + \sin x = 1$, $0 \leq x \leq \pi$, $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$



Find the equations of the tangents to the curve that are parallel to the line $y = x$. You must show clear algebraic working.
Solutions based on graphical or numerical approaches are not acceptable. (9 marks)

Now try this ⟶ Exam question bank Q4, Q7, Q28, Q39, Q43, Q52, Q67, Q77

Selected exam question bank questions

Question 4

The curve C has equation $y = 2x^2 + 2x - e^x$.

a. Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ (2 marks)

b. Find the equation of the tangent to C at the point $(0, -1)$. (3 marks)

Question 7

Given that $x = \dfrac{\ln y}{y}$, $y > 0$,

a. find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ (3 marks)

b. Hence, or otherwise, find the value of $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ at $y = e^{-4}$. (3 marks)

Question 28

A curve has equation $y = \cos 2x - \sin^2 2x$, $-\pi \leq x \leq 0$.

a. Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ and $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ (4 marks)

b. Find the x-coordinates of the stationary points of the curve. (3 marks)

c. Show that a local maximum occurs at $x = -\dfrac{\pi}{2}$ (2 marks)

Question 39

$\mathrm{f}(x) = 2^x + 4^{-x}$. Show that the equation of the tangent to the curve with equation $y = \mathrm{f}(x)$ at the point with x-coordinate -1 is $(15 \ln 2)x + 2y + 15 \ln 2 - 9 = 0$. (6 marks)

Question 43

A curve has equation $6x^2 - 6y^2 - 4xy + 40 = 0$.

a. Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of x and y. (4 marks)

b. Show that line joining the two points on the curve where the gradient is $-\dfrac{3}{5}$ has equation $9x + 2y = 0$. (2 marks)

Question 52

$\mathrm{f}(x) = x^2(2 - 5x)^5$

a. Find $\mathrm{f}'(x)$ and $\mathrm{f}''(x)$. (4 marks)

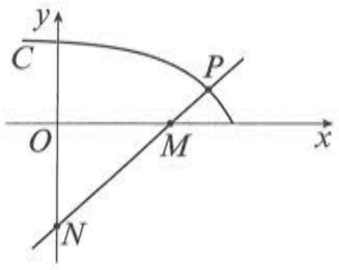
b. Find the x-coordinates of the points where $\mathrm{f}''(x) = 0$. (3 marks)

c. Demonstrate that $x = \dfrac{2}{5}$ is a point of inflection. (2 marks)

Question 67

The graph shows part of the curve C with parametric equations

$$x = 2\cos 2\theta, \quad y = \frac{1}{2}\sin\theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$



a. Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ giving your answer in terms of θ. (4 marks)

P is the point on the curve where $\theta = \dfrac{\pi}{6}$

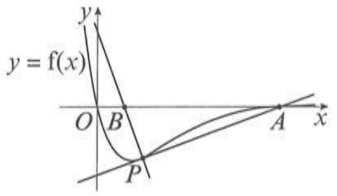
b. Find an equation of the normal to C at P. (4 marks)

The normal to C at P intersects the axes at the points M and N.

c. Find the exact area of triangle OMN. (2 marks)

Question 77

The diagram shows part of the curve with the equation $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = (x^2 - 4x)e^{-x}$.



The point P has coordinates $\left(1, -\dfrac{3}{e}\right)$.

The tangent to the curve at P intersects the x-axis at the point A.

a. Find the equation of the tangent. (4 marks)

The normal to the curve at P intersects the x-axis at the point B.

b. Show that the area of the triangle ABP is $\dfrac{9(e^2+1)}{2e^3}$ (7 marks)

Answers

9.1 Differentiating $\sin x$ and $\cos x$

1.

a. $6\cos x$

b. $-\frac{3}{2}\cos\frac{1}{2}x$

c. $\frac{5}{3}\cos 5x$

d. $-4\sin 4x$

e. $-30\sin 6x$

f. $\frac{7}{2}\sin 7x$

2.

a. $4\cos x + 3\sin x$

b. $\cos 2x - 4\sin x - \frac{3}{2}\cos\frac{1}{4}x$

c. $\frac{15x^{\frac{1}{2}}}{2} + \frac{8}{x^3} - 2\cos\frac{1}{3}x$

d. $-\frac{4}{x^{\frac{3}{2}}} - \frac{1}{2}\sin\frac{1}{6}x$

3. $x = -59.0°,\ 121.0°$

4. $y = 3x - \frac{\pi}{2}$

5. $\left(\frac{\pi}{2}, 9\right),\ \left(\frac{7\pi}{6}, -\frac{9}{2}\right),\ \left(\frac{3\pi}{2}, -3\right),\ \left(\frac{11\pi}{6}, -\frac{9}{2}\right)$

6. $y = -\frac{1}{12}x + \frac{\pi}{12} - 4$

7. $\mathrm{h}\left(\frac{\pi}{2}\right) = \frac{24+\pi^2}{8}$; $\mathrm{h}'(x) = 3\cos x + 4\sin x + x$; $\mathrm{h}'\left(\frac{\pi}{2}\right) = \frac{8+\pi}{2}$

Then use the formula $y - \mathrm{h}\left(\frac{\pi}{2}\right) = \mathrm{h}'\left(\frac{\pi}{2}\right)\left(x - \frac{\pi}{2}\right)$ to find the equation of the tangent.

8. $\mathrm{f}'(x) = \lim_{h\to 0}\frac{\mathrm{f}(x+h) - \mathrm{f}(x)}{h} = \lim_{h\to 0}\frac{\cos(x+h) - \cos x}{h}$

$= \lim_{h\to 0}\frac{\cos x\cos h - \sin x\sin h - \cos x}{h}$

$= \lim_{h\to 0}\left(\frac{\cos h - 1}{h}\cos x - \frac{\sin h}{h}\sin x\right)$

Since $\frac{\cos h - 1}{h} \to 0$ and $\frac{\sin h}{h} \to 1$ the expression inside the limit tends to $(0 \times \cos x - 1 \times \sin x)$

So $\lim_{h\to 0}\frac{\cos(x+h) - \cos x}{h} = -\sin x$

Hence the derivative of $\cos x$ is $-\sin x$.

9.2 Differentiating exponentials and logarithms

1.

a. $-18e^{-3x}$

b. $10e^{2x} + 7e^{-x}$

c. $-\frac{35}{e^{5x}}$

d. $2e^{2x} - 2e^{-2x}$

2.

a. $\frac{1}{x}$

b. $\frac{2}{x}$

c. $-\frac{1}{x}$

d. $\frac{3}{x}$

3.

a. $3^x \ln 3$

b. $5^{2x}(2 \ln 5)$

c. $\left(\frac{2}{5}\right)^{4x} \left(4 \ln \frac{2}{5}\right)$

d. $-\frac{\ln 8}{8^x}$

4. $y = x \ln 2 + \frac{1}{2}(1 + \ln 2)$

5. $-e^{-x} + \frac{6}{x}$

6. $x = 1$, $y = 3e$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 3e - 2$

Rearrange $y - 3e = -\frac{1}{3e-2}(x - 1)$ to obtain

$x + (3e - 2)y - 9e^2 + 6e - 1 = 0$

7. $x = \frac{1}{2}$, $y = \frac{5}{2}$, $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2} \ln \frac{1}{4} + 2 \ln 4 = \ln 8$

Rearrange $y - \frac{5}{2} = (\ln 8)\left(x - \frac{1}{2}\right)$ to obtain

$y = (\ln 8)x + \frac{5}{2} - \frac{1}{2} \ln 8$

8.

a. $\mathrm{f}(x) = 9^{3x} + 3^{6x} = (3^2)^{3x} + 3^{6x} = 3^{6x} + 3^{6x} = 2(3^{6x})$

b. $\mathrm{f}'(x) = 2(3^{6x} \times 6 \times \ln 3) = (4 \ln 3)(3 \cdot 3^{6x}) = (4 \ln 3)(3^{6x+1})$

9.3 The chain rule

1.

a. $-9(4 - 3x)^2$

b. $\frac{5}{2}(6 + 5x)^{-\frac{1}{2}}$

c. $24x^2(5 + 2x^3)^3$

d. $-30x(7 + 3x^2)^{-6}$

e. $\frac{15}{(4 - 5x)^{\frac{3}{2}}}$

f. $\frac{1}{(6 - x)^2}$

g. $\frac{8}{3}x^3(3 + 2x^4)^{-\frac{2}{3}}$

h. $\frac{21}{2}x^2(7x^3 - 4)^{-\frac{1}{2}}$

2.

a. $3\sin^2 x\cos x$

b. $\dfrac{4}{x}$

c. $(-4\sin 4x)e^{\cos 4x}$

d. $\dfrac{2\ln x}{x}$

e. $-\tan x$

f. $-3e^{-3x}\cos(e^{-3x})$

3.

a. $-\dfrac{2}{x}(3-\ln 6x)$

b. $-\dfrac{12}{e^2}$

4. $\dfrac{1}{20}$

5. $x+10y-12=0$

6. $9x-16y-30=0$

7. $Q\left(-\dfrac{5\pi}{6},-\dfrac{3}{4}\right)$, $\dfrac{\mathrm{d}y}{\mathrm{d}x}=10\sin x\cos x$

Simplify $y+\dfrac{3}{4}=\dfrac{5\sqrt{3}}{2}\left(x+\dfrac{5\pi}{6}\right)$ to obtain

$12y-30\sqrt{3}x-25\pi\sqrt{3}+9=0$

8. $A\left(-\sqrt{3},\dfrac{2\pi}{3}\right)$, $\dfrac{\mathrm{d}y}{\mathrm{d}x}=-\dfrac{1}{8\cos 4y}$

Simplify $y-\dfrac{2\pi}{3}=-4(x+\sqrt{3})$ to obtain $4x+y+4\sqrt{3}-\dfrac{2\pi}{3}=0$

9. $(8\ln 2)x+y+8\ln 2-2=0$

9.4 The product rule

1.

a. $(4-5x)^3-15x(4-5x)^2$

b. $9x^2(2x-1)^6+36x^3(2x-1)^5$

c. $\dfrac{96}{x^2}(6x-7)^3-\dfrac{8}{x^3}(6x-7)^4$

d. $16x(2x^3-4)^{-3}-144x^4(2x^3-4)^{-4}$

2.

a. $2x\cos 2x-2x^2\sin 2x$

b. $3e^{3x}\sin 5x+5e^{3x}\cos 5x$

c. $-e^{-x}(2x^2+3)^3+12xe^{-x}(2x^2+3)^2$

d. $-6\cos 4x\cos 2x+12\sin 4x\sin 2x$

e. $x+2x\ln x$

f. $4e^{4x}\ln(\cos 3x)+3e^{4x}\tan 3x$

3. 76

4.

a. $-2xe^{-2x}(x^3-2x^2-3x+3)$

b. $12e^{3x}\cos 2x-8e^{3x}\sin 2x-3e^{3x}\sin x-e^{3x}\cos x$

c. $2x\ln 4x+x-\dfrac{5}{x}\cos\dfrac{1}{2}x+\dfrac{5}{2}(\ln 4x)\sin\dfrac{1}{2}x$

5.

a. $-e^{-x}\cos 3x - 3e^{-x}\sin 3x$

b. $4x^3\ln(\sin x) + x^4\cot x$

6.

a. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = x^3(2-5x^2)^2(-100x^2+16)$
$n=2$, $a=-100$, $b=0$ and $c=16$

b. $x = 0, \pm\sqrt{\dfrac{2}{5}}, \pm\dfrac{2}{5}$

7.

a. $\mathrm{f}'(x) = e^{4x}(4\sin x + \cos x) = 0$
$e^{4x} \neq 0$, so $4\sin x + \cos x = 0 \Rightarrow \tan x = -\dfrac{1}{4}$

b. $\mathrm{f}\left(\dfrac{\pi}{2}\right) = e^{2\pi}$, $\mathrm{f}'\left(\dfrac{\pi}{2}\right) = 4e^{2\pi}$
Therefore the gradient of the normal is $-\dfrac{1}{4e^{2\pi}}$
Simplify $y - e^{2\pi} = -\dfrac{1}{4e^{2\pi}}\left(x - \dfrac{\pi}{2}\right)$ to obtain
$x + 4e^{2\pi}y - 4e^{4\pi} - \dfrac{\pi}{2} = 0$

8. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \sin 2x + 2(x-\pi)\cos 2x$
At $x = \dfrac{3\pi}{4}$, $y = \dfrac{\pi}{4}$ and $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -1$
Simplify $y - \dfrac{\pi}{4} = -1\left(x - \dfrac{3\pi}{4}\right)$ to obtain $x + y = \pi$

9.5 The quotient rule

1.

a. $-\dfrac{42}{(2x-9)^2}$

b. $\dfrac{6x(6-5x)}{(3-5x)^2}$

c. $\dfrac{-8(3x^3+2)^2 + 18x^2(8x-1)(3x^3+2)}{(3x^3+2)^4}$

d. $\dfrac{7x^3+84x-54}{(x^2+6)^{\frac{3}{2}}}$

2.

a. $\dfrac{8x(\cos 2x + x\sin 2x)}{\cos^2 2x}$

b. $\dfrac{\ln x + e^{-x}\ln x + \frac{1}{x}e^{-x} - 1}{(\ln x)^2}$

c. $\dfrac{e^x(\cos x + 2\sin x)}{\cos^3 x}$

d. $\dfrac{e^{-x}\sin 4x - 4(5-e^{-x})\cos 4x}{\sin^2 4x}$

3. $-\dfrac{34}{9}$

4.

a. $\dfrac{-2x^2+14x-6}{e^{2x}}$

b. $y + \dfrac{5}{e^2} = -\dfrac{e^2}{6}(x-1)$

5. $\dfrac{2\sin x(\cos x - \sin x)}{e^{2x}}$

6. $\dfrac{e^{4x}(4x^2 - 22x + 5)}{x^2(x-5)^2}$

7. $\sqrt{2}e^{\frac{\pi}{8}}x + 3y - 1 - \dfrac{3\pi}{8} = 0$

8.

a. $\mathrm{f}'(x) = \dfrac{-2e^x \cos x \sin x - e^x \cos^2 x}{e^{2x}}$
$\mathrm{f}'(x) = 0 \Rightarrow -2e^x \cos x \sin x - e^x \cos^2 x = 0$
Therefore $2\cos x \sin x + \cos^2 x = 0$
Solve to obtain $\cos x = 0$ or $\tan x = -\dfrac{1}{2}$

b. A: $x = -1.57$; B: $x = -0.46$; C: $x = 1.57$; D: $x = 2.68$

9.6 Differentiating trigonometric functions

1.

a. $4\sec^2 4x$

b. $-14\operatorname{cosec}^2 2x$

c. $-5\cot^4 x \operatorname{cosec}^2 x$

d. $-6\tan x \sec^2 x$

2.

a. $-6\operatorname{cosec} 6x \cot 6x$

b. $-20\sec 5x \tan 5x$

c. $-\dfrac{4}{3}(\operatorname{cosec} x)^{\frac{4}{3}} \cot x$

d. $-24\sec^3 4x \tan 4x$

3.

a. $2x\tan 3x + 3x^2 \sec^2 3x$

b. $\dfrac{(x^2+1)\sec x \tan x\, e^{\sec x} - 2xe^{\sec x}}{(x^2+1)^2}$

c. $-2\cot x$

d. $-e^{-x}\cot 4x - 4e^{-x}\operatorname{cosec}^2 4x$

e. $\dfrac{\sec^2 x(2x\ln x \tan x - 1)}{x(\ln x)^2}$

f. $\dfrac{12(4x-1)^2\cot^2 x + 2(4x-1)^3 \cot x \operatorname{cosec}^2 x}{\cot^4 x}$

4.

a. $2\cos 2y$

b. $\dfrac{1}{2\sqrt{1-x^2}}$

5. $y = \operatorname{cosec} x = \dfrac{1}{\sin x}$
$u = 1$, $\dfrac{\mathrm{d}u}{\mathrm{d}x} = 0$; $v = \sin x$, $\dfrac{\mathrm{d}v}{\mathrm{d}x} = \cos x$
$\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\sin x \times 0 - 1 \times \cos x}{\sin^2 x} = -\dfrac{1}{\sin x} \times \dfrac{\cos x}{\sin x} = -\operatorname{cosec} x \cot x$

6.

a. Use the chain rule to find $\mathrm{f}'(x) = \dfrac{-2\operatorname{cosec}^2 2x}{\cot 2x}$ and then use trigonometric identities to obtain $\mathrm{f}'(x) = -4\operatorname{cosec} 4x$

b. $y = \dfrac{1}{4}x - \dfrac{\pi}{32}$

7. $\dfrac{5}{4}$

8.

a. $4\sec 4y \tan 4y$

b. $\dfrac{1}{4x\sqrt{x^2-1}}$

9.7 Parametric differentiation

1.

a. $\dfrac{3t^2}{2}$

b. $\dfrac{t^4}{2}$

c. $\dfrac{e^{4t} + 4te^{4t}}{3}$

d. $\dfrac{5}{4}\sec^3 t$

e. $\dfrac{1}{6e^{2t}}$

f. $\dfrac{1}{6t^3}$

2.

a. $\sin t$

b. $y = \dfrac{\sqrt{2}}{2}x + \dfrac{\sqrt{2}}{2}$

3.

a. $-\left(\dfrac{3t+4}{2}\right)$

b. $2x + 5y + 61 = 0$

4.

a. $4\cot 2\theta$

b. $(1, 4)$ and $(1, -4)$

5.

a. $-4\cos^3\theta$

b. $y = 2x - 7$

6. $y = 2x + 28$

7.

a. $\dfrac{\operatorname{cosec}^2 t}{2\sec 2t \tan 2t}$

b. $m = -\dfrac{\sqrt{3}}{3}$, simplify $y + 2\sqrt{3} = -\dfrac{\sqrt{3}}{3}(x+4)$ to obtain $x\sqrt{3} + 3y + 10\sqrt{3} = 0$

8.

a. $y = -\frac{25}{36}x + \frac{307}{540}$

b. Rearrange parametric equation for x to $t = \frac{x}{x+1}$

Then $y = \frac{1}{1+\left(\frac{x}{x+1}\right)^2} = \frac{(x+1)^2}{(x+1)^2+x^2} = \frac{x^2+2x+1}{2x^2+2x+1}$

9.8 Implicit differentiation

1. $u = y^5$, $\frac{\mathrm{d}u}{\mathrm{d}y} = 5y^4$

$\frac{\mathrm{d}}{\mathrm{d}x}(y^5) = \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{\mathrm{d}u}{\mathrm{d}y} \times \frac{\mathrm{d}y}{\mathrm{d}x} = 5y^4\frac{\mathrm{d}y}{\mathrm{d}x}$

2. $\frac{\mathrm{d}}{\mathrm{d}x}(x^3y^2) = x^3\frac{\mathrm{d}}{\mathrm{d}x}(y^2) + \frac{\mathrm{d}}{\mathrm{d}x}(x^3)y^2$

$= x^3 \times 2y\frac{\mathrm{d}y}{\mathrm{d}x} + 3x^2y^2$

$= 2x^3y\frac{\mathrm{d}y}{\mathrm{d}x} + 3x^2y^2$

3.

a. $\frac{4x^3}{3y^2}$

b. $\frac{6x-10xy}{5x^2-8y}$

c. $\frac{2xy}{10y+3y^2-x^2}$

d. $\frac{4y}{3(x-5)}$

e. $\frac{\sin x}{2\sin y\cos y}$

f. $\frac{2ye^{2x}+e^{-y}}{xe^{-y}-e^{2x}}$

4.

a. $\frac{8xy-2x}{2y-4x^2}$

b. $\frac{10}{9}$

c. $y = \frac{10}{9}x - \frac{29}{9}$

5. $\frac{7}{13}$

6. $\frac{2e^{-2x}-4y}{4x+3e^{-3y}}$

7.

a. $\frac{\sin 2x}{\cos 2y}$

b. $x + 2y = 0$

8. $x + y + \frac{\pi}{12} = 0$

9.9 Using second derivatives

1.

a. $-e^{-x}\cos 2x - 2e^{-x}\sin 2x$

b. $e^{-x}(4\sin 2x - 3\cos 2x)$

2.

a. $16 + 48e^{4x}$

b. $16 \geq 0$ and $48e^{4x} \geq 0$ for all values of x, hence $\mathrm{f}(x)$ is convex for all values of x.

3.

a. $-4\cos 2x$

b. $-\frac{\pi}{2} \leq x \leq -\frac{\pi}{4}$ and $\frac{\pi}{4} \leq x \leq \frac{\pi}{2}$

4.

a. $\mathrm{f}'(x) = (2x-1)^5 + 10x(2x-1)^4$
$\mathrm{f}''(x) = 20(2x-1)^3(6x-1)$

b. $\mathrm{f}''\left(\frac{1}{2}\right) = 20 \times 0^3 \times 2 = 0$
$\mathrm{f}''\left(\frac{1}{6}\right) = 20 \times \left(-\frac{2}{3}\right)^3 \times 0 = 0$

c. $\mathrm{f}''(0.1) = 4.096 > 0 \quad \mathrm{f}''(0.2) = -0.864 < 0$
As there is a sign change, this confirms there is a point of inflection at $x = \frac{1}{6}$

5. $x \geq \frac{8}{3}$

6.

a. $\mathrm{f}'(x) = (2x-5)^4 + 8x(2x-5)^3$
$\mathrm{f}''(x) = 16(2x-5)^3 + 48x(2x-5)^2$

b. $\mathrm{f}''(1) = 0$ and $\mathrm{f}''(0.95) = -15.376 < 0$ and
$\mathrm{f}''(1.05) = 13.456 > 0$
So there is a point of inflection at $x = 1$

7. $0.55 \leq x \leq \frac{\pi}{4}$

8. $x \leq -3 - \sqrt{3}$ and $-3 + \sqrt{3} \leq x \leq 0$

9.10 Rates of change

1.

a. $8x$

b. 24

2. $\frac{3}{4\sqrt{2}}(5\pi - 4)$

3. $-\frac{3e^8}{64}$

4.

a. $4\pi r^2$

b. $12\pi r^2$

5.

a. $\frac{\mathrm{d}P}{\mathrm{d}t} = kP$

b. The value of the constant k would be negative, or $\frac{\mathrm{d}P}{\mathrm{d}t} = -kP$

6. $\frac{\mathrm{d}I}{\mathrm{d}t} = -k\sqrt{I}$

7.

a. £4083.69

b. Decreasing by £482.25 per year

8.

a. Substitute $r = \left(\frac{S}{4\pi}\right)^{\frac{1}{2}}$ into $V = \frac{4}{3}\pi r^3$ and simplify.

b. $\frac{\mathrm{d}V}{\mathrm{d}S} = \frac{1}{4}\sqrt{\frac{S}{\pi}}$; $\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{\mathrm{d}V}{\mathrm{d}S} \times \frac{\mathrm{d}S}{\mathrm{d}t} = \frac{1}{4}\sqrt{\frac{S}{\pi}} \times 4 = \sqrt{\frac{S}{\pi}}$

Problem solving: Set A

Bronze

a. $\frac{\mathrm{d}y}{\mathrm{d}x} = 4x^3 - 20x^2 + 28x + 6$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12x^2 - 40x + 28$

b. $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12x^2 - 40x + 28 = 4(x-1)(3x-7)$

For $1 \leq x \leq \frac{7}{3}$, $x - 1 \geq 0$ and $3x - 7 \leq 0$

Hence $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} \leq 0$ and the curve is concave.

Silver

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 24x \ln x^2 + 40x = 8x(3 \ln x^2 + 5)$

When $x > 0$, $8x \geq 0$. So solve $3 \ln x^2 + 5 \geq 0$, to get $x \geq e^{-\frac{5}{6}}$

Hence $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} \geq 0$ when $x \geq e^{-\frac{5}{6}}$ and the curve is convex.

Gold $a \geq \frac{1}{2}$

Problem solving: Set B

Bronze

a. $\frac{2x + y - 1}{1 - x}$

b. $y = -x + 2$

Silver

a. $(x+1)^2 + (y-3)^2 = 4$; centre $(-1, 3)$ and radius 2

b. $2\sin t - 1 = 0 \Rightarrow t = \frac{\pi}{6}, \frac{5\pi}{6}$

$y = 3 + \sqrt{3}$ or $y = 3 - \sqrt{3}$, therefore $y = 3 + \sqrt{3}$ at B

c. $3 + 2\sqrt{3}$

Gold $y = x + \frac{\pi}{6}$ and $y = x - \frac{7\pi}{6}$

Selected exam question bank answers

Question 4

a. $4x + 2 - e^x$

b. $y = x - 1$

Question 7

a. $\dfrac{y^2}{1 - \ln y}$

b. $\dfrac{1}{5e^8}$

Question 28

a. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -2\sin 2x - 4\sin 2x \cos 2x$

$\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = -4\cos 2x - 8\cos 4x$

b. $x = -\pi, -\dfrac{2\pi}{3}, -\dfrac{\pi}{2}, -\dfrac{\pi}{3}, 0$

c. At $x = -\dfrac{\pi}{2}$, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$ and $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = -4 \leq 0$ which implies a local maximum.

Question 39

$x = -1$, $y = \dfrac{9}{2}$, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -\dfrac{15}{2}\ln 2$

Rearrange $y - \dfrac{9}{2} = \left(-\dfrac{15}{2}\ln 2\right)(x + 1)$ to obtain

$(15\ln 2)x + 2y + 15\ln 2 - 9 = 0$

Question 43

a. $\dfrac{3x - y}{3y + x}$

b. Solve $\dfrac{12x - 4y}{12y + 4x} = -\dfrac{3}{5}$ to obtain $9x + 2y = 0$.

Question 52

a. $\mathrm{f}'(x) = 2x(2 - 5x)^5 - 25x^2(2 - 5x)^4$

$\mathrm{f}''(x) = 2(2 - 5x)^5 - 100x(2 - 5x)^4 + 500x^2(2 - 5x)^3$

b. 0.04, 0.19, 0.40 (2 d.p.)

c. $\mathrm{f}''(0.4) = 0$ and $\mathrm{f}''(0.35) = 0.82\ldots > 0$ and $\mathrm{f}''(0.45) = -1.75\ldots < 0$, so there is a point of inflection at $x = 0.4$.

Question 67

a. $-\dfrac{1}{16}\operatorname{cosec}\theta$

b. $32x - 4y - 31 = 0$

c. $\dfrac{961}{256}$

Question 77

a. $x - ey = 4$

b. $A(4, 0)$, $B\left(\dfrac{e^2 - 3}{e^2}, 0\right)$

Area $= \dfrac{1}{2} \times \dfrac{3}{e} \times \left(4 - \dfrac{e^2 - 3}{e^2}\right)$

This simplifies to $\dfrac{9(e^2 + 1)}{2e^3}$